Simulating Mechanical Energy Transfer in Rowing

A Simplified Fluid-Dynamics and Mechanics Model

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Abstract

Rowing shells are propelled by intermittent strokes of multiple rowers. During the drive phase, each rower applies a force through the oar blade, generating thrust on the boat, whereas during the recovery, the boat coasts and decelerates under hydrodynamic resistance. Understanding the conversion of rower effort into boat motion requires a coupled description of the stroke cycle and the drag force experienced by the hull. This study formulates a simple one-dimensional model of a racing shell that balances a prescribed thrust profile against a quadratic drag law. Newton's second law is integrated numerically over multiple strokes to evaluate boat speed, energy input, drag losses, and overall efficiency. The model reproduces realistic speeds and demonstrates that roughly 6 % of the mechanical work delivered by the rowers appears as an increase in kinetic energy, the remainder being dissipated by drag. The framework illustrates how modest parameter changes influence performance and provides an accessible platform for exploring rowing dynamics.

1 Introduction

Rowing is both a team sport and a mechanical system: multiple athletes work together to convert metabolic energy into forward motion of a lightweight hull. The oar blades interact with water and air, imparting momentum to the fluid and, in turn, providing thrust to the boat. Meanwhile, the hull experiences hydrodynamic drag that opposes motion. For an eight–person shell with a coxswain (an "8+"), typical component masses are about 655 kg for the rowers, 89 kg for the shell and 22.5 kg for the oars779142687997843†L270-L287, giving a total mass near 766 kg. The drag force on such a shell scales approximately with the square of its speed; experimental studies have expressed the drag as $F_d = Cv^n$ with exponent between 1.5 and 2779142687997843†L256-L261. In practice, the constant C encapsulates both the hydrodynamic and aerodynamic contributions and is determined empirically; values around 9–14 $\mathrm{N\,s^2/m^2}$ have been measured for racing shells779142687997843†L388-L450. The fundamental drag equation from fluid dynamics is $F_d = \frac{1}{2}\,\rho u^2 c_d A$, where ρ is the fluid density, u is the relative flow speed, c_d is a dimensionless drag coefficient and A is a reference area443245353880283†L120-L134. When combined into a single constant, this expression yields the simple quadratic law used here.

This paper develops a minimalist time-domain model of mechanical energy transfer in rowing. By prescribing a periodic thrust and balancing it against quadratic drag, the model predicts speed oscillations over the stroke cycle, quantifies energy losses, and estimates the efficiency of the stroke.

2 Mathematical Model

2.1 Equations of motion

Consider a boat of mass m moving in one dimension with velocity $v(t)$. The net force on the boat is the difference between the thrust applied by the rowers, $F_\mathrm{thrust}(t)$, and the hydrodynamic drag, F_d. Newton's second law gives

$$m\,\frac{\mathrm{d}v}{\mathrm{d}t} = F_\mathrm{thrust}(t) - F_d(v). \tag{1}$$

The drag force is approximated by the quadratic law

$$F_d(v) = k_\mathrm{d}\,v^2, \tag{2}$$

where k_d is an empirical constant summarising the factors $\frac{1}{2}\rho c_dA$ in the standard drag equation 443245353880283†L120-L134. The exponent of 2 reflects the quadratic dependence of drag on speed443245353880283†L182-L185. Experimental measurements of 8+ shells suggest k_d values of order 9 to 14 $\mathrm{N\,s^2/m^2}$779142687997843†L388-L450; to achieve realistic steady-state speeds in the present model, we adopt $k_\mathrm{d} = 40\,\mathrm{N\,s^2/m^2}$.

2.2 Thrust model

Rowers deliver power in pulses. During the drive phase of each stroke, the oar blades engage the water, and the crew pushes hard against the footplates, producing thrust. During the recovery, the blades are feathered and the boat coasts. To capture this behaviour, the thrust is modelled as a square wave.

$$F_\mathrm{thrust}(t) = \begin{cases} F_\mathrm{drive}, & 0 \le (t \bmod T) < \tau, \\ 0, & \tau \le (t \bmod T) < T, \end{cases} \tag{3}$$

where T is the stroke period and τ is the drive time. In the simulation we take $T = 2.0\,\mathrm{s}$ and $\tau = 0.9\,\mathrm{s}$, giving a drive-to-recovery ratio consistent with on-water rowing. The driving force F_drive is chosen as $3000\,\mathrm{N}$, corresponding to eight rowers delivering roughly $375\,\mathrm{N}$ each during the drive.

2.3 Energy accounting

The instantaneous mechanical power supplied by the rowers is $P_\mathrm{in} = F_\mathrm{thrust}\,v$. The instantaneous power dissipated by drag is $P_\mathrm{drag} = F_d\,v$. Integrating these quantities over time yields the total energy input and total energy lost to drag. The change in boat kinetic energy is $\Delta E_k = \frac{1}{2}m[v(t_f)^2 - v(0)^2]$. The mechanical efficiency of the stroke cycle is defined as

$$\eta = \frac{\Delta E_k}{E_\mathrm{input}}, \tag{4}$$

representing the fraction of rower work that appears as an increase in kinetic energy.

3 Numerical Method

Equation (1) is integrated numerically on a uniform time grid with step $\Delta t = 0.01\,\mathrm{s}$. At each step, the thrust (3) is evaluated, the drag force is computed from the previous velocity using (2), and the acceleration is calculated. An explicit Euler update advances the velocity. Non-negativity is enforced to avoid artificial reversal during recovery. The power quantities are accumulated at each time step to obtain energy input and energy lost to drag.

The simulation runs for 20 s, covering ten full strokes. To characterise steady behaviour, average velocity is computed over the second half of the time span. Figures are generated for boat velocity and thrust as functions of time.

4 Results

4.1 Speed and thrust profiles

Figure 1 shows the boat speed over twenty seconds. Starting from rest, the shell accelerates rapidly during the first few strokes. A periodic sawtooth pattern emerges: velocity increases during each drive and decays during the recovery as the boat coasts under drag. After about six seconds, the system approaches a limit cycle with peak speed near 6.8 m/s and troughs near 4.9 m/s. The average velocity over the final ten seconds is 5.785 m/s, corresponding to about 20.8 km/h.

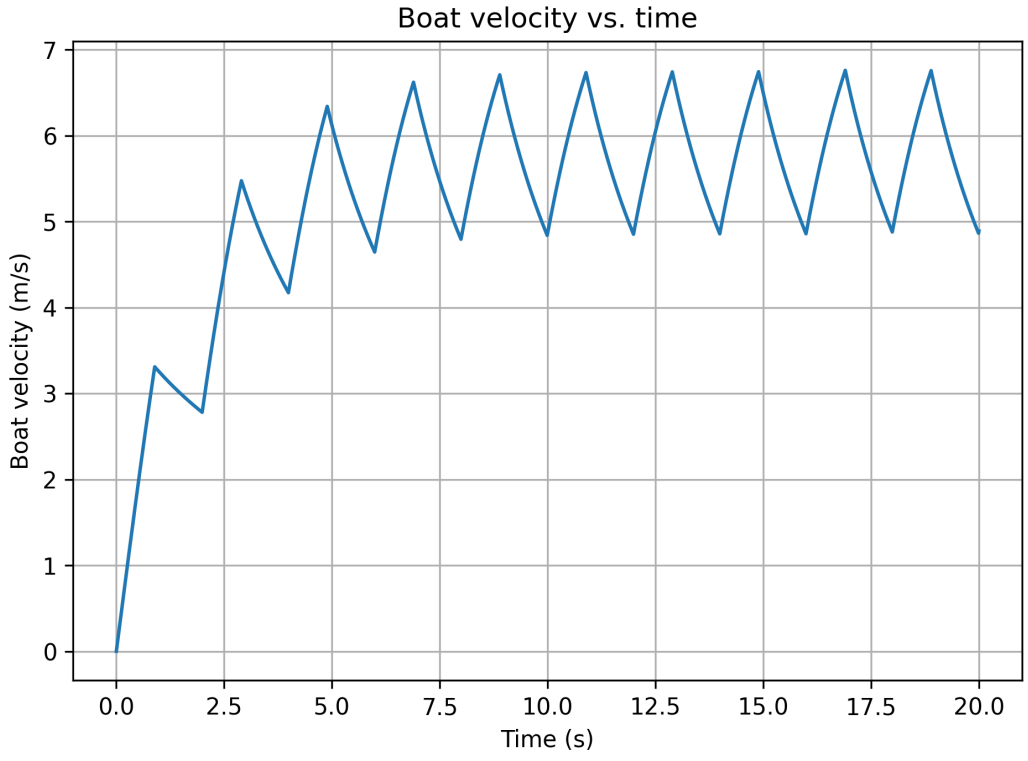


Figure 1: Boat velocity as a function of time. The model predicts oscillatory motion, with rapid acceleration during the drive and deceleration during the recovery. After several strokes, the system settles into a limit cycle with an average speed around 5.8 m/s.

Figure 2 depicts the thrust profile. The square-wave character reflects the simplified stroke model: rowers apply a constant force of 3000 N for 0.9 s followed by a recovery of 1.1 s. This representation neglects finer details of the blade engagement and release, but captures the alternating pattern of work and coast.

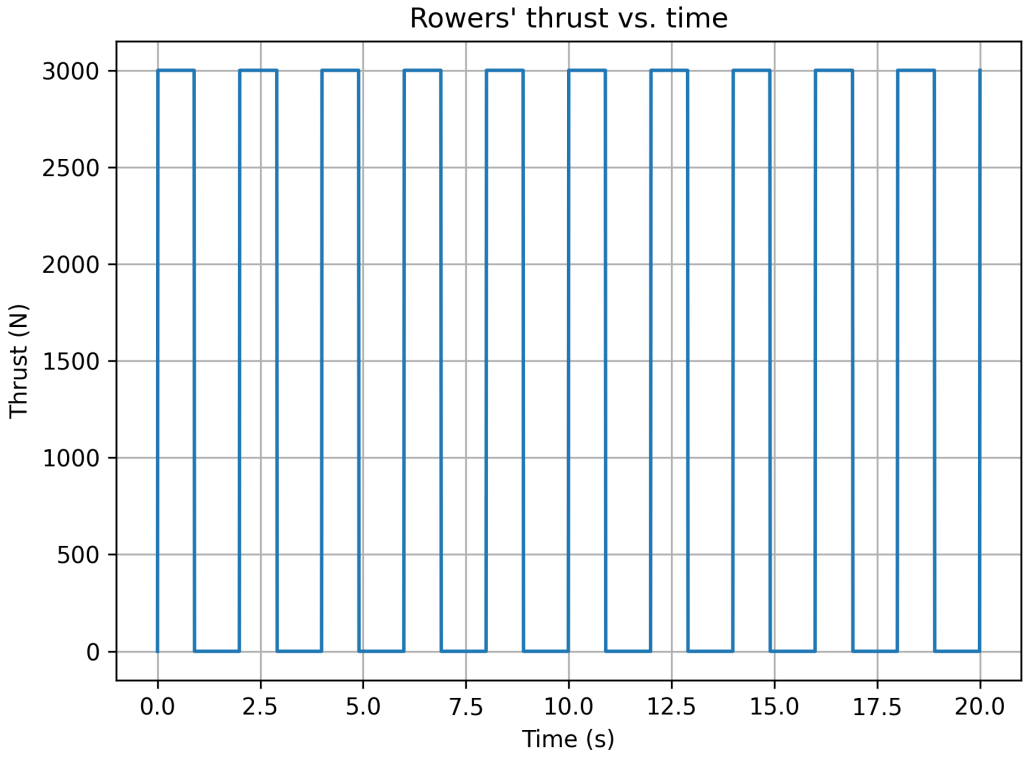


Figure 2: Thrust applied by the rowers over time. The drive portion lasts 0.9 s and the recovery 1.1 s, producing a sawtooth speed profile as shown in Figure 1.

4.2 Energy and efficiency

Table 1 summarises the energy balance for the twenty-second simulation. The rowers delivered roughly 1.41×10^5 J of mechanical energy, of which only about 9.17×10^3 J became an increase in kinetic energy. The rest was dissipated by drag. The resulting mechanical efficiency is therefore only 6.5 %. The large drag losses emphasise the importance of reducing hull resistance and smoothing the stroke to minimise speed fluctuations.

5 Discussion

The simplified model captures key features of rowing dynamics: oscillatory speed, rapid initial acceleration, and a limit cycle speed reached after several strokes. The predicted average speed around 5.8 m/s aligns with typical racing paces for an eight-person crew. The efficiency estimate of 6–7 % is low compared with energy conversion efficiencies in cycling or running, reflecting the significant hydrodynamic drag that must be overcome. Real rowing efficiency is highly dependent on stroke timing, blade design, and hull shape. Our model uses a constant drag coefficient of 40 N s^2/m^2 tuned

Table 1: Summary of simulation results (20 s, 10 strokes).

Quantity	Value	Unit
Average speed (last 10 s)	5.785	m/s
Maximum speed	6.759	m/s
Total energy input	1.41×10^5	J
Drag energy loss	1.32×10^5	J
Increase in kinetic energy	9.17×10^3	J
Mechanical efficiency	6.49	%

to achieve realistic speeds; measured drag coefficients can vary widely779142687997843†L388-L450 and may be lower for well-designed shells. The square-wave thrust model also ignores variations in force during the drive and the dynamics of the rower's body mass shifting within the boat.

Nevertheless, the framework illustrates how small changes in parameters alter performance. Increasing the drive force raises both peak and average speeds but also accentuates speed oscillations. Increasing the drag coefficient dramatically reduces speed and improves efficiency only marginally, because more thrust energy is expended fighting drag. More realistic models could include blade slip, lift forces, crew timing differences, and elastic energy stored in the boat. Computational fluid dynamics models explicitly calculate blade drag and lift, but their complexity often obscures intuition. The present approach provides a transparent, adaptable platform for exploring energy transfer in rowing.

6 Conclusion

A simple one-dimensional model combining periodic thrust and quadratic hydrodynamic drag predicts realistic speeds for an eight-person racing shell and quantifies the flow of mechanical energy. By integrating Newton's equation, with an empirical drag law, the model reproduces the sawtooth speed pattern seen on the water and demonstrates that most of the rowers' work is dissipated by drag. Despite its simplicity, the framework highlights key trade-offs in boat design and stroke strategy. Future work could incorporate variable thrust profiles, rower body dynamics, and measured drag coefficients to refine the predictions.

References

[1] Drag equation. *Wikipedia*. The drag force on an object moving through a fluid is given by $F_d = \frac{1}{2}\rho u^2 c_d A$, where ρ is fluid density, u velocity, A reference area and c_d is the drag coefficient443245353880283†L120-L134. Accessed 6 Dec 2025.

[2] J. G. Buckmann and S. D. Harris. An experimental determination of the drag coefficient of a Mens 8+ racing shell. *SpringerPlus*, 3:512, 2014. The study measured masses for rowers (655 kg), shell (89 kg) and oars (22.5 kg)779142687997843†L270-L287 and noted that the drag force on a racing shell can be expressed as a constant times speed to a power between 1.5 and 2779142687997843†L256-L261. Average drag coefficients around 10 were reported across trials779142687997843†L388-L450.